19. Factors, Multiples & Primes

Year 4 · Mark Scheme — Supplementary Worksheet

Foundation — Answers

1. Not prime — its factors are 1, 2, 3 and 6, so it has more than two factors

2. 1, 2, 7, 14

3. A, B and D (12, 18 and 24 are all multiples of 6; 20 and 28 are not)

4. No (28 ÷ 5 leaves a remainder)

5. 36

6. Prime — its only factors are 1 and itself

7. B — 5 (1, 6 and 8 are all factors of 24; 5 is not)

8. 9, 18, 27, 36, 45

9. No (45 ÷ 10 leaves a remainder)

10. D — 64

11. Not prime — its factors are 1, 3, 5 and 15, so it has more than two factors

12. 1, 2, 3, 6, 9, 18

13. C — 28

14. No (29 ÷ 4 leaves a remainder)

15. C — 81

Standard — Answers

16. 2, 3, 5, 7, 11, 13, 17, 19

17. C — Yes; √16 = 4

18. 1, 2 and 4

19. 12

20. 1×24, 2×12, 3×8, 4×6

21. A and C (37 and 41 are prime; 39 = 3×13, 44 = 4×11 and 49 = 7×7 are not prime)

22. Yes; √64 = 8

23. 1 and 3

24. C — 12

25. 1×40, 2×20, 4×10, 5×8

26. 2, 3, 5, 7, 11, 13

27. B — Yes; √100 = 10

28. 1, 2, 3, 4, 6 and 12

29. C — 20

30. 1×45, 3×15, 5×9

31. Neither (27 = 3 × 9, so it's not prime; and it is not a perfect square)

32. 6 factors: 1, 2, 4, 5, 10, 20

33. B — 8

34. A square number (36 = 6 × 6)

35. B, C and D (5, 6 and 10 are all factors of 30; 4 and 12 are not)

Hard — Answers

36. 36 minutes (the lowest common multiple of 9 and 12)

37. 21, which is odd

38. 10 factors: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48

39. B — 10

40. A square number (49 = 7 × 7), and not prime since it has factors other than 1 and itself

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